



TURBIDITY TEST OF WATER

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D15A INFT

PROJECT OBJECTIVE

- Arduino-based system to measure water turbidity.
- Uses a turbidity sensor to display real-time readings.
- Useful for environmental monitoring and water quality.

COMPONENTS USED

- Arduino UNO R3
- Turbidity Sensor (SEN0189)
- 16x2 LCD Display (with I2C)
- Jumper Wires
- USB Cable

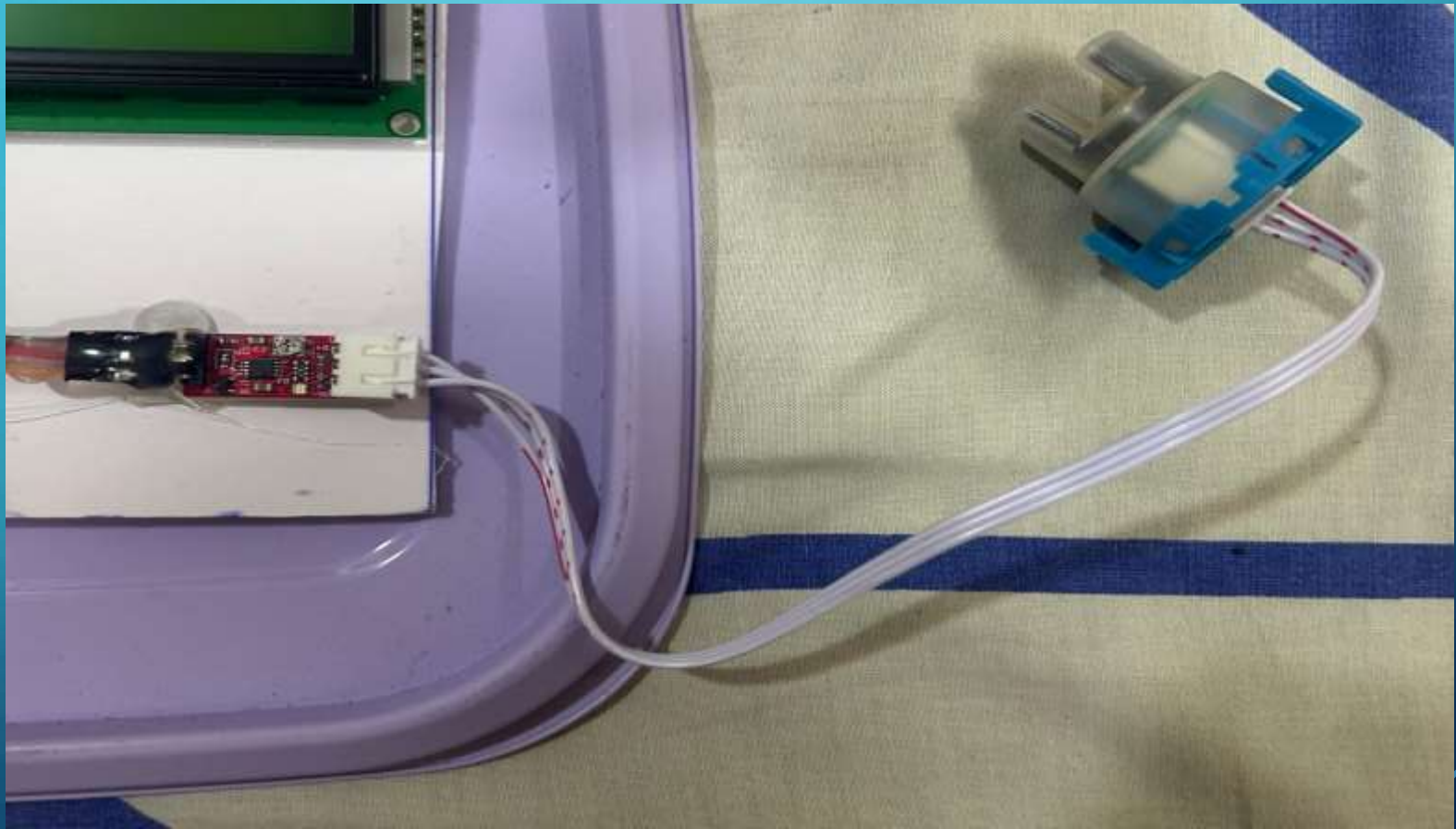
Technical specifications of Components

Arduino UNO	Turbidity Sensor (SEN0189)	16x2 LCD Display (I2C)	Jumper Wires	USB Cable
• 5V operating voltage • 14 digital I/O pins, 6 analog inputs • 16 MHz clock speed • 32 KB flash memory	• 5V supply voltage • 0–1000 NTU measurement range • Analog output signal • Waterproof sensor head	• 5V operating voltage • 16 characters × 2 lines • I2C communication (only 2 pins) • Adjustable contrast	• For quick connections • Male/Male, Male/Female, Female/Female	• USB A to B type • Used for programming and powering Arduino

PIN CONNECTIONS - TURBIDITY SENSOR

- VCC to 5V
- GND to GND
- Signal to A0

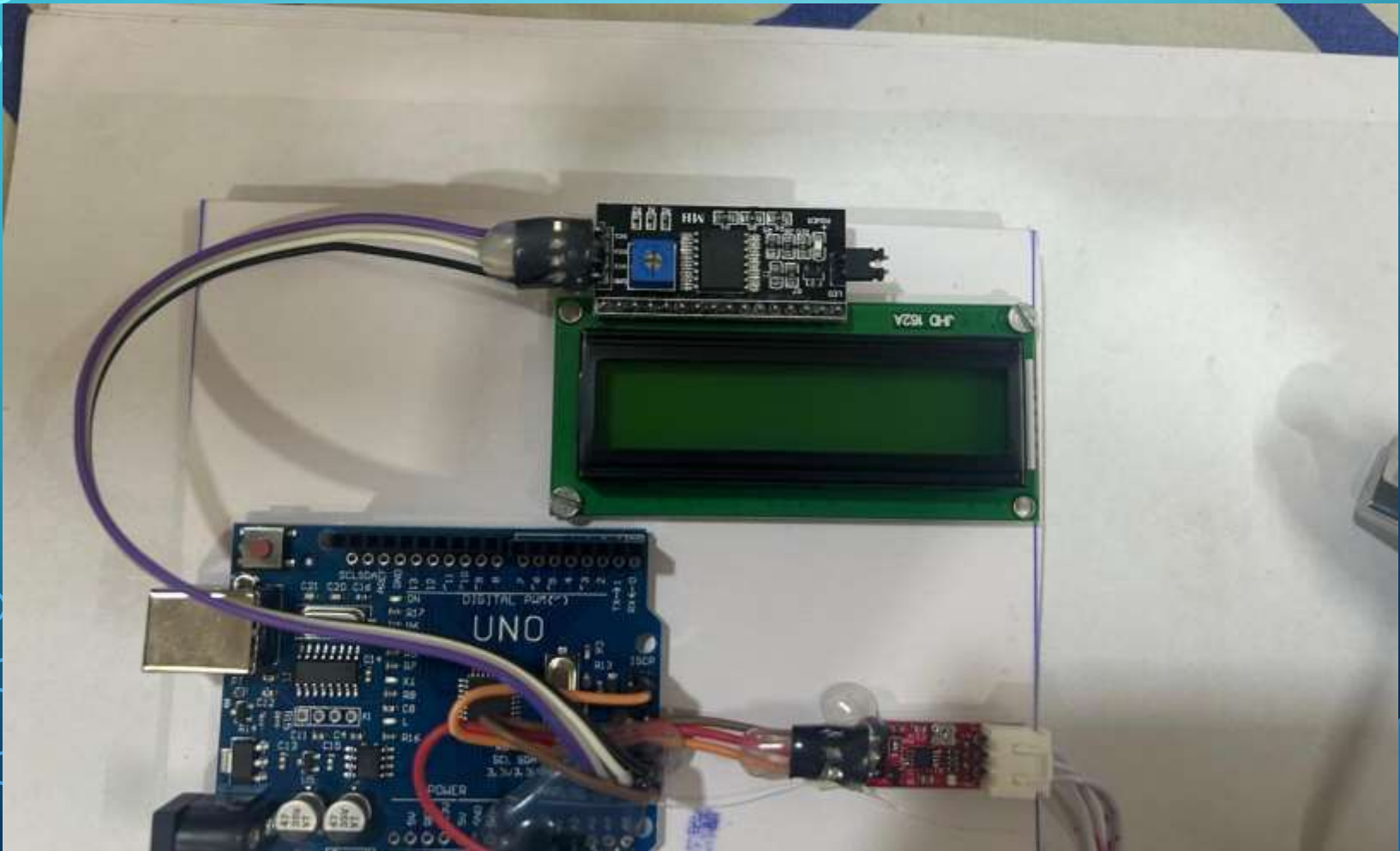
Turbidity Sensor Connection



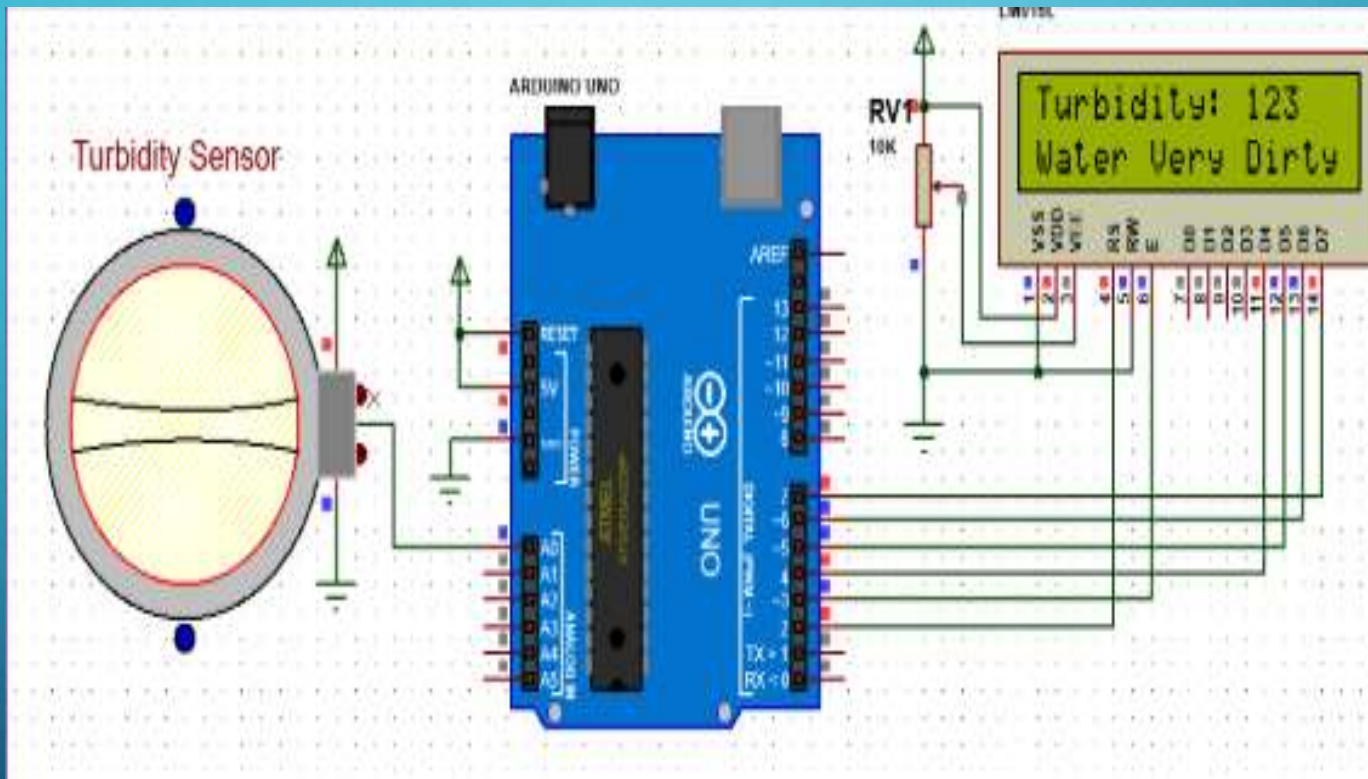
PIN CONNECTIONS - LCD (I2C)

- VCC to 5V
- GND to GND
- SDA to A4
- SCL to A5

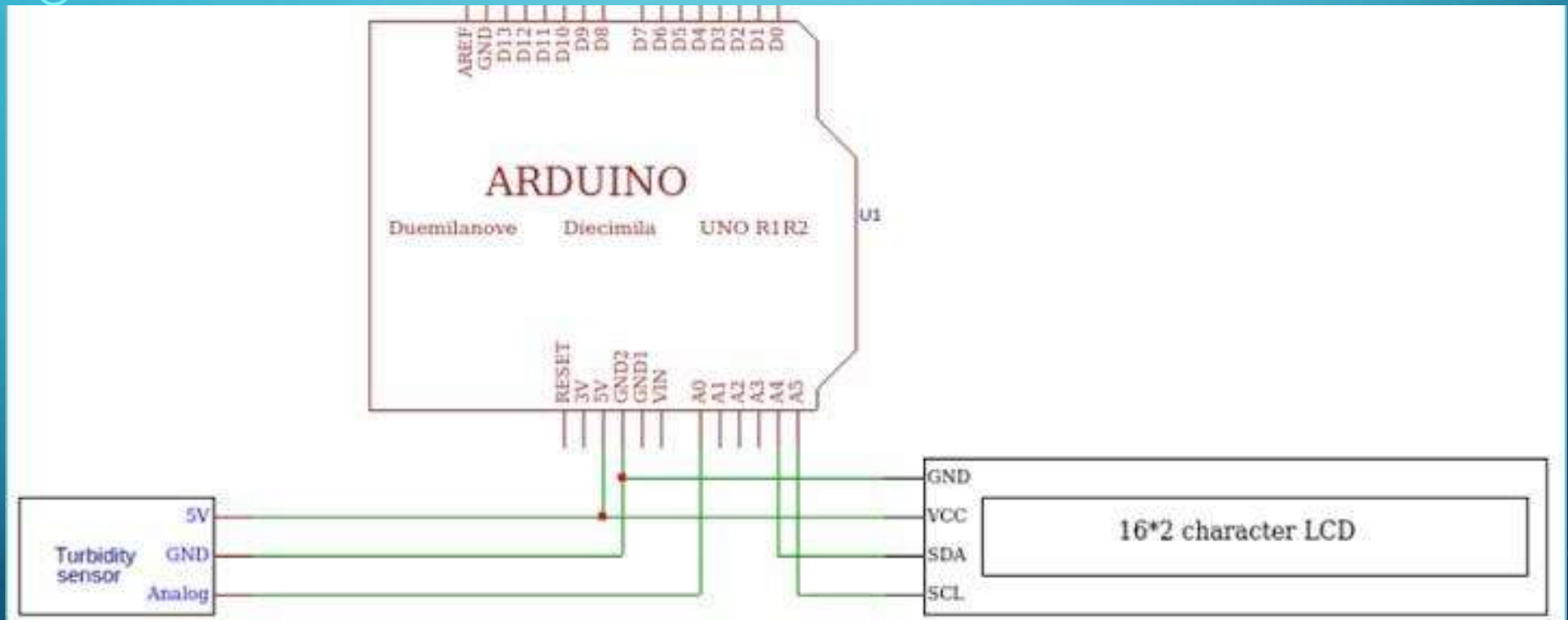
LCD CONNECTIONS WITH ARDUINO



CIRCUIT DIAGRAM



Schematic Diagram



PROCEDURE - TINKERCAD SIMULATION

- Simulate Arduino UNO and verify LCD.
- Connect turbidity sensor.
- Run sample codes.
- Display Turbidity on LCD.

PROCEDURE - PHYSICAL IMPLEMENTATION

- Connect LCD and turbidity sensor.
- Upload and run code.
- Test with various water samples.
- Monitor LCD for turbidity readings.

ARDUINO CODE (PART 1)

```
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 2, 16);

int sensorPin = A0;

void setup()
{
  Serial.begin(9600);
  // initialize LCD
  lcd.init();
  // turn on LCD backlight
  lcd.backlight();
}

void loop() {
  int sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  int turbidity = map(sensorValue, 0, 520, 100, 0);
  delay(100);
  lcd.setCursor(0, 0);
  lcd.print("Turbidity:");
  lcd.print("    ");
  lcd.setCursor(10, 0);
  lcd.print(turbidity);

  delay(100);
  if (turbidity < 20) {
    lcd.setCursor(0, 1);
    lcd.print(" its CLEAR ");
    Serial.print(" its CLEAR ");
  }
  if ((turbidity > 20) && (turbidity < 50)) {
    lcd.setCursor(0, 1);
    lcd.print(" its CLOUDY ");
    Serial.print(" its CLOUDY ");
  }
}
```

ARDUINO CODE (PART 2)

```
delay(100);  
if (turbidity < 20) {  
  lcd.setCursor(0, 1);  
  lcd.print(" its CLEAR ");  
  Serial.print(" its CLEAR ");  
}  
if ((turbidity > 20) && (turbidity < 50)) {  
  lcd.setCursor(0, 1);  
  lcd.print(" its CLOUDY ");  
  Serial.print(" its CLOUDY ");  
}  
if (turbidity > 50) {  
  lcd.setCursor(0, 1);  
  lcd.print(" its DIRTY ");  
  Serial.print(" its DIRTY ");  
}
```

EXPLANATION OF CODE

- Initializes and sets up the LCD.
- Reads sensor's analog values.
- Judges the Purity of Water Sample
- Displays turbidity readings on LCD.

APPLICATIONS

- River and lake pollution monitoring
- Industrial wastewater analysis
- Aquaculture
- Drinking water quality testing

CONCLUSION

- Learned sensor integration with Arduino.
- Displayed and interpreted turbidity levels.
- Built a practical environmental tool.